Geometry

Labix

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Contents

1 Vector Spaces with a Metric

1.1 Isometric Isomorphisms

Definition 1.1.1 (Isometric Isomorphisms)
Let V, W be normed vector spaces. Let $T : V \to W$ be a function. We say that T is an isometric isomorphism if T is an isometry and a linear isomorphism.

2 Affine Geometry

Affine geometry refers to the geometry done on spaces but ignoring the notions of coordinates and measures of angles and distance.

2.1 The Analytical Definition of Affine Space

We must first define a certain notion of space in order to discuss affine geometry. These spaces which has the barebone structure needed to talk about affine geometry is called affine spaces. We would like affine spaces to have the following properties.

- Aside from the underlying set of the space, we would like a set of vectors so that we can add a vector with a given point into a new point.
- Given any two points x, y in the space, there is exactly one vector that sends x to y.

We take the following definition for the underlying space of affine geometry.

Definition 2.1.1 (Affine Space)
An affine space consists of the following data.
- A set A.
- A vector space V.
- A free and transitive action of V on A.

The action of the vector space on the points give us addition of points and vectors, and the free and transitive action guarantees a unique vector connecting any two points.

Lemma 2.1.2
Let $(A, V, +)$ be an affine space. Let $v \in V$. Then the function $A \to A$ defined by $a \mapsto a + v$ is a bijection.

Definition 2.1.3 (Subtraction)
Let $(A, V, +)$ be an affine space. Let $a, b \in A$. Define the subtraction of a from b to be the unique vector $b - a \in V$ such that
$$a + (b - a) = b$$

Definition 2.1.4 (Affine Subspaces)
Let $(A, V, +)$ be an affine space. An affine subspace of A is a set of the form
$$a + W = \{a + w \mid w \in W\}$$
where $W \subseteq V$ is a subspace.

2.2 Affine Transformations

Definition 2.2.1 (Affine Transformations)
Let $(A, V, +)$ and $(B, W, +)$ be affine spaces. Let $f : A \to B$ be a function. We say that f is an affine transformation if the induced function $\overline{f} : V \to W$ defined by
$$q - p \mapsto f(q) - f(p)$$
is a well defined linear map for $p, q \in A$.

Definition 2.2.2 (Affine Isomorphism)
Let $(A, V, +)$ and $(B, W, +)$ be affine spaces. Let $f : A \to B$ be an affine transformation. We say that f is an affine isomorphism if $\overline{f} : V \to W$ is an isomorphism of vector spaces.

Proposition 2.2.3
Let $(A, V, +)$ be an affine space. Let $a \in A$. Then there is a unique affine isomorphism $f : A \to V$ such that $f(a) = 0_V$.

2.3 The Synthetic Approach

3 Euclidean Geometry

3.1 Lines and Incidences

Definition 3.1.1 (Euclidean Space)
A Euclidean space is an affine space (E, V) such that V is a finite dimensional inner product space over $\mathbb{R}$.

In Linear Algebra, we classified that up to isomorphism, every finite dimensional inner product space is isomorphic to $\mathbb{R}^n$ equipped with the standard dot product. Since we have also seen that every affine space has a unique isomorphism to its underlying vector space by specifying the point of origin, we lose no generality by working with $\mathbb{R}^n$ and the standard dot product. Therefore some literature define Euclidean space to be $\mathbb{R}^n$ equipped with its standard dot product.

Definition 3.1.2 (The Metric of Euclidean Spaces)
Let E be a Euclidean space. Define the metric $d : E \times E \to \mathbb{R}$ on E to be

$$d(P, Q) = \left|\overrightarrow{PQ}\right|$$

where $\overrightarrow{PQ}$ is the unique vector such that $P + \overrightarrow{PQ} = Q$.

Lemma 3.1.3
Let E be a Euclidean space. Then $d : E \times E \to \mathbb{R}$ is indeed a metric on E.

Definition 3.1.4 (Line Segments)
Let (E, d) be an Euclidean space. Let $P, Q \in E$. Define the line from P to Q to be

$$\overline{PQ} = \{R \in E \mid d(P, R) + d(R, Q) = d(P, Q)\}$$

If $E = \mathbb{R}^n$ instead, another way to write the set of points that form the line segment between two points P and Q is

$$\overline{PQ} = \{P + t(Q - P) \mid 0 \le t \le 1\}$$

Definition 3.1.5 (Lines)
Let (E, d) be an Euclidean space. Let $P, Q \in E$. Define the line from P to Q to be

$$L_{P,Q} = \left\{ R \in E \;\middle|\; \begin{array}{c} d(P,R)+d(R,Q)=d(P,Q) \text{ or} \\ d(P,Q)+d(Q,R)=d(P,R) \text{ or} \\ d(P,Q)+d(P,R)=d(Q,R) \end{array} \right\}$$

Similarly, if $E = \mathbb{R}^n$ then there is an equality

$$L_{P,Q} = \{P + t(Q - P) \mid t \in \mathbb{R}\}$$

Definition 3.1.6 (Collinearity)
Let E be an Euclidean space. Let $P, Q, R \in E$. We say that P, Q, R are collinear if there exists a line $L \subseteq E$ such that

$$P, Q, R \in L$$

Lemma 3.1.7
Let E be an Euclidean space. Let $T : E \to E$ be an isometry. If $P, Q, R \in E$ are collinear, then $T(P), T(Q), T(R)$ are collinear.

Proof
Let $P, Q, R \in E$ be collinear. Without loss of generality suppose that $d(P, Q) + d(Q, R) = d(P, R)$. Since T is an isometry, we have

$$d(T(P), T(Q)) + d(T(Q), T(R)) = d(T(P), T(R))$$

Hence $T(R) \in L_{T(P),T(Q)}$ and so $T(P), T(Q), T(R)$ are collinear. ■

Lemma 3.1.8
Let E be an Euclidean space. Let $T : E \to E$ be an isometry. Let $L \subseteq E$ be a line. Then $T(L)$ is also a line.

Proof
I claim that $T(L) = L_{T(P),T(Q)}$ for any $P, Q \in L$. Let $W \in T(L)$. Then there exists $R \in L$ such that $W = T(R)$. Since $R \in L = L_{P,Q}$, P, Q, R are collinear. Hence $T(P), T(Q), T(R)$ are collinear. Thus $T(R) \in L_{T(P),T(Q)}$. Hence $T(L) \subseteq L_{T(P),T(Q)}$. Conversely, suppose that $W \in L_{T(P),T(Q)}$, then $T(P), T(Q), W$ are collinear. Since T is an isometry, there exists $R \in E$ such that $W = T(R)$. Moreover, T^{-1} is also an isometry hence $T(P), T(Q), T(R)$ collinear implies that P, Q, R are collinear. Hence $R \in L_{P,Q} = L$. Hence $R \in T(L)$ and so $T(L) = L_{P,Q}$. Hence $T(L)$ is a line.
■

Definition 3.1.9 (Parallel Lines)

Proposition 3.1.10 (The Parallel Postulate is True in Euclidean Geometry)

Definition 3.1.11 (Perpendicular Lines)

3.2 Angles

Recall that from real analysis, we defined the power series

$$\cos(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k}}{(2k)!} \quad \text{and} \quad \sin(x) = \sum_{k=0}^{\infty} \frac{(-1)^k x^{2k+1}}{(2k+1)!}$$

We have seen that $\cos(x)$ is strictly decreasing in the interval $[0, \pi]$ and so $\cos(x) = k$ for $-1 \leq k \leq 1$ has a unique solution between 0 and π.

Definition 3.2.1 (Angles of a Euclidean Space)
Let E be a Euclidean space. Let $P, Q, R \in E$ be points. Define the angle between $\overline{PQ}$ and $\overline{QR}$ to be the unique real number $\angle PQR \in \mathbb{R}$ such that

$$\cos(\angle PQR) = \frac{\overrightarrow{QP} \cdot \overrightarrow{QR}}{\left|\overrightarrow{QP}\right|\left|\overrightarrow{QR}\right|}$$

and $0 \leq \angle PQR \leq \pi$.

Lemma 3.2.2
Let E be a Euclidean space. Let $P, Q, R \in E$ be points. If $Q \in \overline{PR}$, then $\angle PQR = \pi$.

Proof
Since $Q \in \overline{PR}$, we have that $d(P, Q) + d(Q, R) = d(P, R)$, which translates to $\left|\overrightarrow{PQ}\right| + \left|\overrightarrow{QR}\right| = \left|\overrightarrow{PR}\right|$.
Hence the vectors $\overrightarrow{PQ}$ and $\overrightarrow{QR}$ are collinear in $\mathbb{R}^n$ as a Euclidean space. Thus there exists $\lambda > 0$

such that $-\lambda\overrightarrow{QP} = \overrightarrow{QR}$. Then we have

$$\begin{aligned}\cos(\angle PQR) &= \frac{\overrightarrow{QP}\cdot\overrightarrow{QR}}{\left|\overrightarrow{QP}\right|\left|\overrightarrow{QR}\right|} \\ &= \frac{-\lambda\overrightarrow{QP}\cdot\overrightarrow{QP}}{\lambda\left|\overrightarrow{QP}\right|\left|\overrightarrow{QP}\right|} \\ &= \frac{-\lambda\left|\overrightarrow{QP}\right|^2}{\lambda\left|\overrightarrow{QP}\right|^2} \\ &= -1\end{aligned}$$

Since $\cos(x)$ is bijective in $[0,\pi]$ and $\cos(\pi) = -1$, we conclude that $\angle PQR = \pi$. ■

Proposition 3.2.3 (Vertically Opposite Angles)
Let E be a Euclidean space. Let $P, Q, R, S \in E$ such that $\overline{PQ}$ meets $\overline{RS}$ at the unique point $T \in E$. Then we have $\angle PTR = \angle QTS$.

3.3 Classifying Isometries of Euclidean Space

Definition 3.3.1 (Affine Maps) A map $T : \mathbb{R}^n \to \mathbb{R}^m$ is affine if it is of the form

$$T(x) = Ax + b$$

for all $x \in \mathbb{R}^n$ for some linear map A and $b \in \mathbb{R}^k$.

Proposition 3.3.2 Let $n, k \in \mathbb{N} \setminus \{0\}$. Let $T : \mathbb{R}^n \to \mathbb{R}^k$ be a function. Then the following are equivalent.

- T is an affine map.
- The map $L(x) = T(x) - T(0)$ is a linear map.
- For all $\lambda \in \mathbb{R}$, $T(\lambda x + (1-\lambda)y) = \lambda T(x) + (1-\lambda)T(y)$

Proof Let $T : \mathbb{R}^n \to \mathbb{R}^k$ be a map.

- (1) $\iff$ (2): Define $L(x) = T(x) - T(0)$. Then T is affine if and only if L is linear, if and only if $L(\lambda x + \mu y) = \lambda L(x) + \mu L(y)$ if and only if $T(\lambda x + \mu y) - T(0) = \lambda(T(x) - T(0)) + \mu(T(y) - T(0))$.
- (2) $\implies$ (3): This is obtained by setting $\mu = 1 - \lambda$.
- (3) $\implies$ (2): We have

$$\begin{aligned}\lambda x + \mu y &= \frac{1}{2}(2\lambda x) + \frac{1}{2}(2\mu y) \\ T(\lambda x + \mu y) &= \frac{1}{2}T(2\lambda x) + \frac{1}{2}T(2\mu y) && \text{(By (3))}\end{aligned}$$

Also by (3), we have

$$\begin{aligned}2\lambda x &= 2\lambda x + (1-2\lambda)0 \\ T(2\lambda x) &= 2\lambda T(x) + (1-2\lambda)T(0)\end{aligned}$$

And similarly,

$$T(2\mu y) = 2\mu T(y) + (1-2\mu)T(0)$$

Combining the three, we have

$$\begin{aligned}
T(\lambda x+\mu y) &= \frac{1}{2}T(2\lambda x)+\frac{1}{2}T(2\mu y) \\
&= \frac{1}{2}(2\lambda T(x)+(1-2\lambda)T(0))+\frac{1}{2}(2\mu T(y)+(1-2\mu)T(0)) \\
&= \lambda T(x)+\left(\frac{1}{2}-\lambda\right)T(0)+\mu T(y)+\left(\frac{1}{2}-\lambda\right)T(0) \\
T(\lambda x+\mu y)-T(0) &= \lambda(T(x)-T(0))+\mu(T(y)-T(0))
\end{aligned}$$

■

Proposition 3.3.3 Let $n \in \mathbb{N} \setminus \{0\}$. Let $L : \mathbb{R}^n \to \mathbb{R}^n$ be a linear map. Let A be the matrix representing T under the standard basis. Then the following are equivalent.

- L is an isometry
- $\|L(x)\| = \|x\|$ for all $x \in \mathbb{R}^n$
- $\langle L(x), L(y)\rangle = \langle x, y\rangle$ for all $x, y \in \mathbb{R}^n$
- A is an orthogonal matrix.

Proof Let $L : \mathbb{R}^n \to \mathbb{R}^k$ be linear.

- $(1) \implies (2)$: This is trivial since isometries are distance preserving
- $(2) \implies (3)$: This is given by the polarization identity
- $(3) \implies (4)$: We have that

$$\begin{aligned}
\left(A^T A\right)_{ij} &= \sum_{k=1}^n (A^T)_{ik} A_{kj} \\
&= \sum_{k=1}^n A_{ki} A_{kj} \\
&= \sum_{k=1}^n L(e_i)_k L(e_j)_k \\
&= \langle L(e_i), L(e_j)\rangle \\
&= \langle e_i, e_j\rangle && \text{(By (3))} \\
&= \delta_{ij}
\end{aligned}$$

  Thus $A^T A = I$

- $(4) \implies (1)$: Suppose that $A^T A = I$. Then A is invertible and thus is a bijection. Thus we just need to show that $d(x, y) = d(L(x), L(y))$.

$$\begin{aligned}
\|L(x)\|^2 &= \langle L(x), L(x)\rangle \\
&= (L(x))^T L(x) \\
&= x^T A^T A x \\
&= x^T x \\
&= \langle x, x\rangle \\
&= \|x\|^2
\end{aligned}$$

  Thus $d(L(x), L(y)) = \|L(x) - L(y)\| = \|L(x-y)\| = \|x-y\| = d(x, y)$.

■

Proposition 3.3.4 Let $n \in \mathbb{N} \setminus \{0\}$. Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be a function. Then T is an isometry if and only if T is affine and given by $T(x) = Ax + b$, where A is orthogonal.

Proof We have shown that Euclidean Isometries are line preserving, thus by 1.3.2 they are affine and in the form $T(x) = Ax + b$. We now check that $G(x) = Ax = T(x) - b$ is an isometry. G is distance preserving since

$$\begin{aligned} d(G(x), G(y)) &= \|G(x) - G(y)\| \\ &= \|T(x) - b - T(y) + b\| \\ &= \|T(x) - T(y)\| \\ &= \|x - y\| && (T \text{ is an isometry}) \\ &= d(x, y) \end{aligned}$$

We now show that G is bijective. Since T is an isometry, T is bijective and $T(x) - b$ is also bijective with $T^{-1}(x) = x + b$ and thus G is an isometry. By the above proposition, A is orthogonal. ■

Proposition 3.3.5 Let $\phi : O(n, \mathbb{R}) \to \text{Aut}(\mathbb{R}^n)$ be the function defined by $\phi(A)(x) = Ax$ for $x \in \mathbb{R}^n$. Then ϕ is a group homomorphism. Moreover, there is a group isomorphism

$$\psi : \text{Isom}(\mathbb{R}^n, d) \to \mathbb{R}^n \rtimes_\phi O(n, \mathbb{R})$$

given by $T \mapsto (T(0), T - T(0))$.

Theorem 3.3.6 (Canonical Form of Orthogonal Matrices) Let $M \in O(n, \mathbb{R})$ be an orthogonal matrix. Then there exists an orthonormal basis of $\mathbb{R}^n$ such that M is similar to a matrix of the form

$$\begin{pmatrix} I_k & & & & \\ & -I_m & & & \\ & & B_1 & & \\ & & & \ddots & \\ & & & & B_l \end{pmatrix}$$

where

$$B_i = \begin{pmatrix} \cos(\theta_i) & -\sin(\theta_i) \\ \sin(\theta_i) & \cos(\theta_i) \end{pmatrix}$$

Proof Let $L : \mathbb{R}^n \to \mathbb{R}^n$ be an isometry. We prove the result by induction on n. Extend the domain of L such that it becomes a map from $\mathbb{C}^n \to \mathbb{C}^n$. By the fundamental theorem of algebra, $c_A(L)$ has at least one root, λ. We know that $|\lambda| = 1$. There are two cases.

- If $\lambda \in \mathbb{R}$, then $\lambda = \pm 1$. Choose an eigenvector $z = x + iy$, with $x, y \in \mathbb{R}^n$. We have that $\lambda(x + iy) = L(x + iy) = L(x) + iL(y)$, thus x, y respectively are both real eigenvectors of λ. At least one of these x, y is non zero, else $z = 0$. Now $\frac{x}{\|x\|}$ is a basis for $W = \mathbb{C} \cdot x$ and the matrix for L is
$$\begin{pmatrix} \pm 1 & 0 \\ 0 & L|_{W^\perp} \end{pmatrix}$$
- If $\lambda \in \mathbb{C} \setminus \mathbb{R}$, choose an eigenvector z. We must have $\overline{\lambda}$ is also an eigenvalue, with eigenvector $\overline{z}$. Let $W = E_\lambda \oplus E_{\overline{\lambda}}$. By the above lemma, there exists a real orthonormal basis for W. In terms of a basis for $W \oplus W^\perp$, the matrix for L is
$$\begin{pmatrix} \begin{pmatrix} \cos(\theta) & -\sin(\theta) \\ \sin(\theta) & \cos(\theta) \end{pmatrix} & 0 \\ 0 & L|_{W^\perp} \end{pmatrix}$$

By the above, in both cases, $L|_{W^\perp}$ is an isometry of $W^\perp$, so we can apply the induction step to $W^\perp$. Eventually, we will have decomposed V into mutually orthogonal subspaces $W_1, \dots, W_M$, all invariant under L. Either W_i has dimension 2 with a real orthonormal basis with respect to which the matrix of $L|_W$ has the form as in the above lemma, or W_i has dimenison 1, with normalized real basis, and L acts as 1 or -1. Since the W_i are mutally orthogonal, the basis consisting of the union of all these basis is orthonormal. By reordering the subspaces appropriately we obtain the above form. ■

I would say that the main part of this theorem is that you essentially decompose $\mathbb{R}^n$ into $W_1 \oplus \cdots \oplus W_n$, where they are mutually orthogonal. It might be the case that an isometry in $\mathbb{R}^2$ or $\mathbb{R}^3$ allows some orthogonal to rotate or reflect, leaving the rest invariant. Thus performing each rotation or reflection in $\mathbb{R}^3$ is only transforming one of $W_1, \dots, W_n$.

Definition 3.3.7 (Hyperplane)
Let $n \in \mathbb{N} \setminus \{0\}$. A hyperplane in $\mathbb{R}^n$ is a set of the form

$$\Pi = \{v + b \mid v \in V\}$$

for $V \subseteq \mathbb{R}^n$ a vector subspace of dimension $n-1$.

Definition 3.3.8 (Fixed Points) Let $T : \mathbb{R}^n \to \mathbb{R}^n$ be an isometry. Denote the set of fixed points of T to be

$$\text{Fix}(T) = \{x \in \mathbb{R}^n \mid T(x) = x\}$$

Definition 3.3.9 (Reflection on Hyperplane)
Let $\Pi \subseteq \mathbb{R}^n$ be a hyperplane. A reflection in Π is an isometry $T : \mathbb{R}^n \to \mathbb{R}^n$ such that

$$\text{Fix}(T) = \Pi$$

Proposition 3.3.10 The reflection on hyperplane exists and is unique.

Proof Let $\Pi = V + b$. Pick $v \in V^\perp$. Take a basis of $\mathbb{R}^n$ consisting of v together with a basis for V. With respect to the basis, define a linear map T with matrix

$$A = \begin{pmatrix} -1 & 0 \\ 0 & I_{n-1} \end{pmatrix}$$

Note that this fixes V and is not an identity. Let $S(x) = x - b$. I claim that $\rho = S^{-1} \circ T \circ S$ is a reflection. I prove that $\text{Fix}(\rho) = \Pi$. This means solving $x = Ax + (I - A)b$. We have that $(A - I)(x - b) = 0$. Solving this gives

$$x = \begin{pmatrix} 0 + b_1 \\ t_2 + b_2 \\ \ddots \\ t_n + b_n \end{pmatrix}$$

But then $x \in \Pi$ since x is of the form $b + v$ with $v \in V$. Thus ρ_Π exists.
Let ρ' that is not the identity mapping fixes Π. Then $R = T \circ \rho' \circ T^{-1}$ by 1.4.2 fixes V and fixes $V^\perp$. Thus $R(v) = \lambda v$ for some $\lambda \in \mathbb{R}$. Since R is an isometry, $|\lambda| = 1$. Since R is not the identity, $\lambda = -1$ and thus R has matrix A. Thus $\rho = \rho'$ and we have proved uniqueness. ■

Lemma 3.3.11 Let $p \in \mathbb{R}^n$. Let $V \subseteq \mathbb{R}^n$ and $v \perp V$. Let ρ_Π be the reflection on a hyperplane $\Pi = V + b$. Then

$$\rho_\Pi(p) = p - 2\langle p - b, v\rangle v$$

Proof Let d be the shortest Euclidean distance between P and the hyperplane. To obtain the reflection of P along the hyperplane, we calculate the vector that starts at P with magnitude d and direction towards the hyperplane, and perpendicular to it, then add it to P two times to obtain it.

Let the point where the tail of v is on Π be Q. Note that $QP = P - b$. Since $\|v\| = 1$, we have that the magnitude of the projection of QP on to v is given by $\langle P - b, v\rangle$. Thus using our method, we obtain that $\rho_\Pi(P) = P - 2\langle P - b, v\rangle v$. ■

Lemma 3.3.12 Let $p, q \in \mathbb{R}^n$ be distinct. Then there exists a reflection ρ with $\rho(p) = q$.

Proof Let $v = \frac{p-q}{\|p-q\|}$. Let $b = \frac{p+q}{2}$. I claim that $\Pi = \mathbb{R}v^\perp + b$ admits a reflection such that $\rho_\Pi(p) = q$.

$$\begin{aligned}\rho_\Pi(p) &= p - 2\left\langle p - \frac{p+q}{2}, \frac{p-q}{\|p-q\|}\right\rangle \frac{p-q}{\|p-q\|} \\ &= p - \langle p - q, p - q\rangle \frac{p-q}{\|p-q\|^2} \\ &= q\end{aligned}$$

Thus we are done. ■

Theorem 3.3.13 Let $T \in \text{Isom}(\mathbb{R}^n, d)$ be an isometry of $\mathbb{R}^n$. Then T is the composition of at most $n+1$ reflections.

Proof Let T be an isometry. Let $P = T(0)$ and $Q = 0$. Then by the above, there exists a reflection R such that $R(T(0)) = 0$. Thus $L = R \circ T$ is a linear isometry. Choose an orthonormal basis $v_1, \dots, v_n$ so that by the normal form theorem, the matrix M has the following form

$$M = \begin{pmatrix} I_k & & & & \\ & -I_m & & & \\ & & B_1 & & \\ & & & \ddots & \\ & & & & B_l \end{pmatrix}$$

Denote J_i to be the $n \times n$ identity matrix, except -1 is at the (i, i)th spot. Note that this is the matrix of a reflection in $(\mathbb{R}v_i)^\perp$. Let B_θ be the matrix of rotation of degree θ. Then it can be decomposed into

$$B_i = A_i \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \begin{pmatrix} \cos(\theta_i) & \sin(\theta_i) \\ \sin(\theta_i) & -\cos(\theta_i) \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

The matrix A_θ has eigenvalues 1 and -1, with corresponding eigenvectors $\left(\cos\left(\frac{\theta_i}{2}\right), \sin\left(\frac{\theta_i}{2}\right)\right)$ and $\left(\sin\left(\frac{\theta_i}{2}\right), -\cos\left(\frac{\theta_i}{2}\right)\right)$ respectively.

Define $C_i = J_{k+m+2i}$ and

$$D_i = \begin{pmatrix} I_{k+m+2i-2} & 0 & 0 \\ 0 & B_i & 0 \\ 0 & 0 & I_{n-k-m-2i} \end{pmatrix}$$

and

$$E_i = \begin{pmatrix} I_{k+m+2i-2} & 0 & 0 \\ 0 & A_i & 0 \\ 0 & 0 & I_{n-k-m-2i} \end{pmatrix}$$

Note that E_i is a reflection in the plane orthogonal to $\sin\left(\frac{\theta_i}{2}\right) v_{k+m+2i-1} - \cos\left(\frac{\theta_i}{2}\right) v_{k+m+2i}$. Then we have

$$M = J_{k+1} \circ \cdots \circ J_{k+m} \circ D_1 \circ \cdots \circ D_l$$

and thus

$$M = J_{k+1} \circ \cdots \circ J_{k+m} \circ E_1 \circ C_1 \circ \cdots \circ E_l \circ C_l$$

which is the product of at most n reflections. Since $T = R^{-1} \circ L$, T is the product of at most $n+1$ reflections. ■

Corollary 3.3.14
Let $n \in \mathbb{N} \setminus \{0\}$. Then we have

$$\text{Isom}(\mathbb{R}^n, d) = \langle T : \mathbb{R}^n \to \mathbb{R}^n \mid T \text{ is a reflection } \rangle$$

Proof
The above theorem implies that every isometry is the composition of finitely many reflections. ■

3.4 Theorems of Triangles

Definition 3.4.1 (Triangles)
Let E be a Euclidean space. A triangle in E consists of three distinct points $P, Q, R \in E$ together with the line segments

$$\triangle PQR = \overline{PQ} \cup \overline{PR} \cup \overline{QR}$$

Theorem 3.4.2 (Angle Sum of Triangle)
Let E be a Euclidean space. Let $\triangle P, Q, R \subseteq E$ be a triangle. Then we have

$$\angle PQR + \angle QRP + \angle RPQ = \pi$$

Definition 3.4.3 (Isosceles Triangle)
Let E be a Euclidean space. Let $\triangle PQR \subseteq E$ be a triangle. We say that $\triangle PQR$ is isosceles if $\overline{PQ} = \overline{PR}$ (up to a permutation of P, Q, R).

Proposition 3.4.4
Let E be a Euclidean space. Let $\triangle P, Q, R \subseteq E$ be a triangle. Then the following are equivalent.
- $\triangle PQR$ is an isosceles triangle with $\overline{PQ} = \overline{PR}$.
- $\angle PQR = \angle PRQ$.
- There exists an isometry $\phi : E \to E$ such that $\phi(P) = P$, $\phi(Q) = R$, $\phi(R) = Q$.

3.5 Theorems of Circles

4 Spherical Geometry

4.1 The Metric Structure on Spheres

Definition 4.1.1 (n-Dimensional Sphere) Let $r \geq 0$. Define the n dimensional sphere of radius r to be

$$S_r^n = \{(x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} : \|x\| = r\}$$

Definition 4.1.2 (The Spherical Metric)
Define the spherical metric on $S^n(r)$ to be the function $d_{S_r^n} : S^n(r) \times S^n(r) \to \mathbb{R}$ given by

$$d_S(P, Q) = r \cdot \angle POQ = r\cos^{-1}\left(\frac{\langle P, Q\rangle}{r^2}\right)$$

Proposition 4.1.3
(S_r^n, d_S) is a metric space.

Note: We lose no generality assuming that $r = 1$.

4.2 Lines and Incidences in Spherical Geometry

Definition 4.2.1 (Lines in Spherical Geometry)
Let $P, Q \in S_r^n$. Define the line from P to Q to be

$$\overline{PQ}^S = \{R \in S_r^n \mid d_S(P, R) + d_S(R, Q) = d_S(P, Q)\}$$

Definition 4.2.2 (Great Circles)
A great circle in S_r^n is a subset $C \subseteq S_r^n$ of the form

$$C = S_r^n \cap T$$

where $T \subseteq \mathbb{R}^{n+1}$ is a vector subspace of dimension 2.

Proposition 4.2.3
Let $L \subseteq S_r^n$ be a subset. Then L is a line if and only if L is a great circle in S_r^n.

Definition 4.2.4 (Antipodal Points)

Proposition 4.2.5
Antipodal iff intersection of two great circles

Proposition 4.2.6
No parallel lines: any two lines intersect in antipodal points

4.3 Angles

Definition 4.3.1 (Angles)
Let $P, Q, R \in S_r^n$. Let $f, g : [0, 1] \to S_r^n$ be componentwise differentiable functions such that $f(0) = g(0) = Q$ and $f(1) = P$, $g(1) = R$. Define the spherical angle between $\overline{QP}^S$ and $\overline{QR}^S$ to be the unique real number $\angle_S PQR \in \mathbb{R}$ such that

$$\cos(\angle_S PQR) = \frac{f'(0) \cdot g'(0)}{|f'(0)||g'(0)|}$$

and $0 \leq \angle_S PQR \leq \pi$.

Check: Angles in $\mathbb{R}^n$ have a similar definition.

Proposition 4.3.2
Let $P, Q, R \in S_r^n$. Let $O = (0, \dots, 0) \in \mathbb{R}^{n+1}$ be the origin. Let n_1, n_2 be normal vectors to the vector subspaces $\mathbb{R}\langle P, Q\rangle$ and $\mathbb{R}\langle Q, R\rangle$ respectively. Let $S, T \in \mathbb{R}^{n+1}$ be points in the intersection of $\mathbb{R}\langle P, Q\rangle$ and $\mathbb{R}\langle Q, R\rangle$ with $n_1 \times n_2$ respectively. Then we have

$$\angle_S PQR = \angle SOT$$

Proposition 4.3.3
Let $P, Q, R \in S^n$. Let $O = (0, \dots, 0) \in \mathbb{R}^{n+1}$ be the origin. Then we have

$$\cos(\angle_S QPR) = \cos(\angle_S PRQ)\cos(\angle_S PQR) + \sin(\angle_S PRQ)\sin(\angle_S PQR)\cos(\angle QOR)$$

4.4 The Group of Isometries of the n-Sphere

Proposition 4.4.1
There is a group isomorphism

$$\text{Isom}(S_r^n, d_S) \cong O(n+1)$$

4.5 Geometry of the 2-Sphere

Definition 4.5.1 (Triangles)

Theorem 4.5.2 (Girad's Theorem) Let $\triangle PQR$ be a spherical triangle on S^2. With P, Q, R distinct points and such that the interior of $\triangle PQR$ does not intersect the great circles which contain the sides of $\triangle PQR$. Then the area of the triangle is equal to $\angle PQR + \angle PRQ + \angle QPR - \pi$.

The reason that we have this much constraints is for us to make sure that we are talking about the same triangle which should be the smallest triangle among the many triangles created from the great circles (or at least those without a segment of a great circle in it).

5 Hyperbolic Geometry

5.1 The Lorentz Inner Product

Definition 5.1.1 (Lorentz Inner Product)
Define the Lorentz inner product on $\mathbb{R}^{n+1}$ to be the function $\langle -, - \rangle_L : \mathbb{R}^{n+1} \times \mathbb{R}^{n+1} \to \mathbb{R}$ given by

$$\langle (x_1, \dots, x_{n+1}), (y_1, \dots, y_{n+1}) \rangle = -x_1 y_1 + \sum_{i=2}^{n+1} x_i y_i$$

Definition 5.1.2 (Lorentz Norm)
Define the Lorentz norm on $\mathbb{R}^{n+1}$ to be the function $\| - \|_L : \mathbb{R}^{n+1} \to \mathbb{R}$ given by

$$\|x\|_L = \sqrt{\langle x, x \rangle_L}$$

Lemma 5.1.3 (Hyperbolic Polarization Identity)
Let $x, y \in \mathbb{R}^{n+1}$. Then we have

$$\langle x, y \rangle_L = \frac{1}{4} \left(\|x + y\|_L^2 - \|x - y\|_L^2 \right)$$

Definition 5.1.4 (Lorentz Identity Matrix)
Define the Lorentz identity matrix to be

$$J_n = \begin{pmatrix} -1 & 0 \\ 0 & I_{n-1} \end{pmatrix} \in M_n(\mathbb{R})$$

Lemma 5.1.5
Let $x, y \in \mathbb{R}^{n+1}$. Then we have

$$\langle x, y \rangle_L = x^T J_{n+1} y = \langle Jx, y \rangle$$

Definition 5.1.6 (Lorentz Orthonormal Vectors)
Let $v_1, \dots, v_{n+1} \in \mathbb{R}^{n+1}$. We say that the vectors are Lorentz orthonormal if

$$\langle v_i, v_j \rangle_L = \begin{cases} 1 & \text{if } 2 \leq i = j \leq n+1 \\ -1 & \text{if } i = j = 1 \\ 0 & \text{otherwise} \end{cases}$$

Lemma 5.1.7
Let $v_1, \dots, v_{n+1} \in \mathbb{R}^{n+1}$ be a set of Lorentz orthonormal vectors. Then $v_1, \dots, v_{n+1}$ forms a basis for $\mathbb{R}^{n+1}$.

Definition 5.1.8 (Physical Interpretation of Vectors)
Let $x \in \mathbb{R}^{n+1}$.

- We say that x is space-like if $\|x\|_L > 0$.
- We say that x is time-like if $\|x\|_L \in \mathbb{C} \setminus \mathbb{R}$.
- We say that x is light-like if $\|x\|_L = 0$.

5.2 Hyperbolic Space

Definition 5.2.1 (Hyperbolic Space)
Define the n-dimensional hyperbolic space to be

$$H^n = \{x = (x_1, \dots, x_{n+1}) \in \mathbb{R}^{n+1} \mid \|x\|_L = i \text{ and } x_1 > 0\}$$

Definition 5.2.2 (Hyperbolic Metric)
Define the hyperbolic metric to be the function $d_H : H^n \times H^n \to \mathbb{R}$ given by

$$d_H(x, y) = \cosh^{-1}(-\langle x, y\rangle_L)$$

5.3 Lines and Incidences in Hyperbolic Geometry

Definition 5.3.1 (Physical Interpretation of Subspaces)
Let $V \subseteq \mathbb{R}^{n+1}$ be a vector subspace.

- We say that V is space-like if every non-zero vector in V is space-like.
- We say that V is time-like if V contains at least one time-like vector.
- We say that V is light-like otherwise.

Definition 5.3.2 (Lorentz Planes)
Let $V \subseteq \mathbb{R}^{n+1}$ be a vector subspace of dimension 2. We say that V is a Lorentz plane if V is time-like.

Definition 5.3.3 (Hyperbolic Lines)
A hyperbolic line in H^n is a subset $L \subseteq H^n$ of the form

$$L = V \cap H^n$$

where V is a Lorentz plane.

Lemma 5.3.4
Any two points has a unique line passing through both

5.4 Hyperbolic Angles

Definition 5.4.1 (Hyperbolic Angles)
Let $P, Q, R \in H^n$. Define the hyperbolic angle at between $\overline{QP}^H$ and $\overline{QR}^H$ to be the unique real number $\angle_H PQR \in \mathbb{R}$ such that

$$\cos(\angle_H PQR) = \frac{P \times_L Q, Q \times_L R}{\|P \times_L Q\|_L \|Q \times_L R\|_L}$$

and $0 \leq \angle_H PQR \leq \pi$.

Proposition 5.4.2
Let $x, y, z \in H^n$ be distinct. Let $\alpha = d_{H^n}(z, y)$, $\beta = d_{H^n}(x, z)$, $\gamma = d_{H^n}(x, y)$ and a be the hyperbolic angle of $\triangle xyz$ at x. Then

$$\cosh(\alpha) = \cosh(\beta) \cdot \cosh(\gamma) - \sinh(\beta) \cdot \sinh(\gamma) \cdot \cos(a)$$

Proof WLOG we can assume x, y, z takes the form $x = (1, 0, \dots, 0)$, $y = (\cosh(t_1), \sinh(t_1), 0, \dots, 0)$ and $z = (\cosh(t_2), \cos(\theta)\sinh(t_2), \sin(\theta)\sinh(t_2), 0, \dots, 0)$. This is because Loretnz transformation preserves the Loretnz inner product and thus the distance function remains unchanged. Since the trailing zeroes make no contributions to the calculations, we assume we are working on H^2. Now note that

$$x \times_L \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} = \begin{pmatrix} 0 \\ -a_3 \\ a_2 \end{pmatrix}$$

Thus we have

$$\begin{aligned} \cos(a) &= \frac{\langle x \times_L y, x \times_L z \rangle_L}{\|x \times_L y\|_L \|x \times_L z\|_L} \\ &= \frac{1}{|\sinh(t_1)||\sinh(t_2)|} \left\langle \begin{pmatrix} 0 \\ 0 \\ \sinh(t_1) \end{pmatrix}, \begin{pmatrix} 0 \\ -\sin(\theta)\sinh(t_2) \\ \cos(\theta)\sinh(t_2) \end{pmatrix} \right\rangle_L \\ &= \cos(\theta) \end{aligned}$$

Now we have that

$$\begin{aligned} \cosh(\gamma) &= -\langle x, y \rangle_L = \cosh(t_1) \\ \cosh(\beta) &= -\langle x, z \rangle_L = \cosh(t_2) \\ \cosh(\alpha) &= -\langle z, y \rangle_L = \cosh(t_1)\cosh(t_2) - \cos(\theta)\sinh(t_1)\sinh(t_2) \end{aligned}$$

Combining these and using the fact that $\cos(a) = \cos(\theta)$ gives our result. ■

5.5 The Group of Isometries of Hyperbolic Space

Definition 5.5.1 (Lorentz Transformation)
Let $T : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ be a function. We say that T is a Lorentz transformation if

$$\langle T(x), T(y) \rangle_L = \langle x, y \rangle_L$$

for all $x, y \in \mathbb{R}^{n+1}$.

Definition 5.5.2 (Lorentz Orthogonal Matrix)
Let $A \in M_{n+1}(\mathbb{R})$ be a matrix. We say that A is Lorentz orthogonal if $A^T J_{n+1} A = J_{n+1}$.

Proposition 5.5.3
Let $T : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ be a linear map. Then the following are equivalent.
- T is a Lorentz transformation.
- $[T]_E^E$ is Lorentz orthogonal, where E is the standard basis of $\mathbb{R}^{n+1}$.
- $\|T(x)\|_L = \|x\|_L$ for all $x \in \mathbb{R}^{n+1}$.
- T is linear and $T(e_1), \dots, T(e_{n+1})$ forms a Lorentz orthonormal basis for $\mathbb{R}^{n+1}$.

Proposition 5.5.4
Let $A \in M_{n+1}(\mathbb{R})$ be a matrix. Then the following are equivalent.
- A is Lorentz orthogonal.
- The map $x \mapsto Ax$ is a Lorentz transformation.
- $\|Ax\|_L = \|x\|_L$ for all $x \in \mathbb{R}^{n+1}$.

Definition 5.5.5 (Lorentz Group)
Define the Lorentz group to be

$$O(1, n) = \{A \in M_{n+1}(\mathbb{R}) \mid A \text{ is Lorentz orthogonal } \}$$

Proposition 5.5.6
There is a one to one bijection

$$O(1,n) \overset{1:1}{\longleftrightarrow} \{T \in \mathrm{Hom}_{\mathbb{R}}(\mathbb{R}^{n+1}, \mathbb{R}^{n+1}) \mid T \text{ is a Lorentz transformation } \}$$

given by $A \mapsto (x \mapsto Ax)$.

Definition 5.5.7 (Positive Lorentz Transformation)
Let $T : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ be a function. We say that T is a positive Lorentz transformation if the following are true.

- T is a Lorentz transformation.
- $x \in \mathbb{R}^{n+1}$ is time-like if and only if $T(x)$ is time-like.

Definition 5.5.8 (Positive Lorentz Orthogonal Matrix)
Let $A \in M_{n+1}(\mathbb{R})$ be a matrix. We say that A is a positive Lorentz orthogonal matrix if the following are true.

- A is a Lorentz orthogonal.
- If $A = (a_{i,j})_{1 \leq i,j \leq n+1}$, then $a_{1,1} > 0$.

Lemma 5.5.9
Let $T : \mathbb{R}^{n+1} \to \mathbb{R}^{n+1}$ be a linear map. Then T is a positive Lorentz transformation if and only if $[T]_E^E$ is a positive Lorentz orthogonal matrix, where E is the standard basis for $\mathbb{R}^{n+1}$.

Lemma 5.5.10
Let $A \in M_{n+1}(\mathbb{R})$ be a matrix. Then the following are equivalent.

- A is a positive Lorentz orthogonal matrix.
- x is a time-like vector if and only if Ax is a time-like vector.
- The map $x \mapsto Ax$ is a positive Lorentz transformation.

Definition 5.5.11 (Positive Lorentz Group)
Define the positive Lorentz group to be

$$O^+(1,n) = \{A \in O(1,n) \mid A \text{ is a positive Lorentz orthogonal matrix } \}$$

Proposition 5.5.12
There is a one to one bijection

$$O^+(1,n) \overset{1:1}{\longleftrightarrow} \{T \in \mathrm{Hom}_{\mathbb{R}}(\mathbb{R}^{n+1}, \mathbb{R}^{n+1}) \mid T \text{ is a positive Lorentz transformation } \}$$

given by $A \mapsto (x \mapsto Ax)$.

Lemma 5.5.13 For every set of linearly independent $a, b, c \in H^n$, there exists an element $A \in O^+(1,n)$ such that

$$\begin{aligned} Aa &= (1, 0, \dots, 0) \\ Ab &= (b_1, b_2, 0, \dots, 0) \text{ with } b_1 > 0 \\ Ac &= (c_1, c_2, c_3, 0 \dots, 0) \text{ with } c_1 > 0 \end{aligned}$$

Proof Let $a, b, c, z_4, \dots, z_{n+1}$ be a basis for H^n. We construct a Lorentz orthonormal basis for $\mathbb{R}^{n+1}$. Since $a \in H^n$ implies $\|a\| = i$, we can choose $v_1 = a$. We find the basis by induction, where

v_k is constructed by $a, b, c, z_4, \ldots, z_k$. Now let $w_2 = b + \langle v_1, b\rangle_L v_1$. Observe that

$$\begin{aligned}\langle w_2, v_1\rangle_L &= \langle b, v_1\rangle_L + \langle v_1, b\rangle_L \langle v_1, v_1\rangle_L \\ &= \langle b, v_1\rangle_L - \langle v_1, b\rangle_L && (\langle v_1, v_1\rangle_L = -1) \\ &= 0\end{aligned}$$

Thus w_2 is Lorentz orthogonal to v_1. Now just choose $v_2 = \frac{w_2}{\|w_2\|_L}$. Note that this is similar to the Gram-schimdt procedure in Euclidean Space. In general, suppose that $v_1, \ldots, v_k$ are now made orthogonal. Define

$$w_{k+1} = a - \sum_{i=1}^{k} \langle z_{k+1}, v_i\rangle_L v_i$$

Then w_{k+1} will be orthogonal to all of $v_1, \ldots, v_k$ thus we can choose $v_{k+1} = \frac{w_{k+1}}{\|w_{k+1}\|}$. Thus a Lorentz orthonormal basis $v_1, \ldots, v_{n+1}$ is formed.
With respect to this basis, we have the desired results. ■

Proposition 5.5.14
There is a group isomorphism

$$\text{Isom}(H^n) = O^+(1, n)$$

5.6 Geometry of 2-Dimensional Hyperbolic Space

Definition 5.6.1 (Lorentz Cross Product)
Let $x, y \in \mathbb{R}^3$. Define the Lorentz cross product of x and y to be

$$x \times_L y = J_3 x \times y$$

Lemma 5.6.2 (Hyperbolic Binet-Cauchy Identity)
Let $x, y, z, w \in \mathbb{R}^3$. Then we have

$$\langle x \times_L y, z \times_L w\rangle_L = -\langle x, z\rangle_L \langle y, w\rangle_L + \langle x, w\rangle_L \langle y, z\rangle_L$$

Lemma 5.6.3
Let $x, y \in \mathbb{R}^3$. Then we have

$$x \times_L y = Jy \times Jx$$

5.7 ???

Lemma 5.7.1 If $x \in \mathbb{R}^n$ with $\langle x, x\rangle_L < 0$, and $w \in \mathbb{R}^n \setminus \{0\}$ with $\langle x, w\rangle_L = 0$, then $\langle w, w\rangle_L > 0$. In other words, any vector that is Lorentz orthogonal with a time-like vector must be a space-like vector.

Proof Let $x' = x - x_1 e_1$ and $w' = w - w_1 e_1$. Then $\langle x', x'\rangle_L \geq 0$ and $\langle x, x\rangle_L = -x_1^2 + \langle x', x'\rangle_L$. This means that $x_1 \neq 0$. If $w_1 = 0$, then since $w \neq 0$, we must have $\langle w, w\rangle_L = \langle w', w'\rangle_L > 0$ and we are done.
Now suppose that $w_1 \neq 0$. Then $x_1 w - w_1 x$ has zero e_1 component. I claim that this vector is non-zero. Indeed if $x_1 w = w_1 x$, then

$$0 = \frac{x_1}{w_1}\langle x, w\rangle_L = \langle x, \frac{x_1}{w_1} w\rangle_L = \langle x, x\rangle_L < 0$$

Now we have that

$$\begin{aligned}0 \leq \|x_1 w - w_1 x\|_L^2 &= \langle x_1 w - w_1 x, x_1 w - w_1 x\rangle_L \\ &= x_1^2 \langle w, w\rangle_L + w_1^2 \langle x, x\rangle_L\end{aligned}$$

Bu this means that

$$\langle w, w\rangle_L \geq -\frac{w_1^2}{x_1^2}\langle x, x\rangle_L > 0$$

and we are done. ■

In layman terms, this means that the orthogonal subspace of a time-like vector is a subspace with all vectors being space-like.

Definition 5.7.2 Let L_1, L_2 be two distinct lines in H^2 with Lorentz plane Π_1 and Π_2 respectively. Let $v \in \Pi_1 \cap \Pi_2 \setminus \{0\}$. Let $V = \Pi_1 \cap \Pi_2$.

- If V is time like, L_1 and L_2 intersect at a point x and $V = \mathbb{R}v = \Pi_1 \cap \Pi_2$
- If V is space like, L_1 and L_2 are parallel and diverge
- If V is light like, L_1 and L_2 are ultraparallel

The following theorem shows that Euclid's Postulate does not hold in hyperbolic space.

Theorem 5.7.3 Let $x \in H^2$. Let L be a line in H^2. Then there are infinitely many lines in H^2 which pass through x and do not intersect L.

Proposition 5.7.4 The hyperbolic angles is invariant under hyperbolic isometries.

Proposition 5.7.5 Let $\triangle PQR$ be a hyperbolic triangle. Let a, b, c be the angles at P, Q, R respectively. Then

$$a + b + c = \pi - \text{Area}(\triangle PQR)$$

Lecture did not have time to thoroughly prove things and so do I. Every Lorentz Transformation must have one of the following four forms:
Lorentz Translation:

$$\begin{pmatrix} \cosh(\beta) & \sinh(\beta) & 0 \\ \sinh(\beta) & \cosh(\beta) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

This transformation does not have a fixed point in H^2. Orientation is preserved.
Lorentz Glide:

$$\begin{pmatrix} \cosh(\beta) & \sinh(\beta) & 0 \\ \sinh(\beta) & \cosh(\beta) & 0 \\ 0 & 0 & -1 \end{pmatrix}$$

This transformation does not have a fixed point in H^2. Orientation is also reversed
Rotation about the point $(1, 0, 0)$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & \cosh(\beta) & \sinh(\beta) \\ 0 & \sinh(\beta) & \cosh(\beta) \end{pmatrix}$$

This transformation has one fixed point, namely $(1, 0, 0)$ and is orientation preserving.
Reflection along $x_2 = 0$:

$$\begin{pmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

This transformation fixes the line $x_2 = 0$ and is orientation reversing.

6 Projective Geometry

6.1 Projective Space

Definition 6.1.1 (Projective Space)
Let $\mathbb{F}$ be a field. Define the n dimensional projective space over $\mathbb{F}$ to be

$$\mathbb{FP}^n = \frac{\mathbb{F}^{n+1} \setminus \{0\}}{\sim}$$

where $x \sim y$ if $\lambda x = y$ for some $\lambda \in \mathbb{F} \setminus \{0\}$.

Note: check that the relation is indeed an equivalence relation.

Proposition 6.1.2
Let $\mathbb{F}$ be a field. There is a one-to-one correspondence

$$\mathbb{FP}^n \quad \overset{1:1}{\longleftrightarrow} \{V \subseteq \mathbb{F}^{n+1} \mid V \text{ is a } 1-dimensional subspace of \mathbb{F}^{n+1}\}$$

Definition 6.1.3 (The Standard Cover of Projective Space)
Let $\mathbb{F}$ be a field. Define

$$U_i = \{[x_0 : \cdots : x_n] \mid x_i \neq 0\}$$

6.2 Projective Linear Subspaces

Definition 6.2.1 (Projective Linear Subspaces) Let $U \subseteq V$ be a vector subspace. Define the projective linear subspace to be

$$\mathbb{P}(U) = \frac{U \setminus \{0\}}{\sim} \subseteq \mathbb{P}(V)$$

with its dimension defined to be $\dim(\mathbb{P}(U)) = \dim(U) - 1$. We define $\dim(\emptyset) = -1$ for conventions.

Definition 6.2.2 (Classification of Subspaces) Let $U \subseteq V$ be a vector subspace where $\dim(V) = n$. We say that $\mathbb{P}(U)$ is

- A single point if $\dim(U) = 1$, or $\dim(\mathbb{P}(U)) = 0$
- A line if $\dim(U) = 2$, or $\dim(\mathbb{P}(U)) = 1$
- A plane if $\dim(U) = 3$, or $\dim(\mathbb{P}(U)) = 2$
- A hyperplane if $\dim(U) = n - 1$, or $\dim(\mathbb{P}(U)) = n - 2$

In particular, note that lines in $\mathbb{R}^{n+1}$ become points. Let $x \in \mathbb{R}^{n+1}$ be fixed. Take the line $y = tx$ for $t \in \mathbb{R}$ as an example. Observe that in the projective space. $x \sim y$ if and only if $y = tx$ for some $t \in \mathbb{R}$. Naturally $y = tx$ becomes a point.

Definition 6.2.3 (Projective Cone and Span) Let V be a vector subspace and let $U \subset \mathbb{P}(V)$ be a subset. Define the projective cone of U to be

$$\tilde{U} = \bigcup_{v \in U} v$$

Define the span of U to be

$$\langle U \rangle = \mathbb{P}(\text{span}(\tilde{U}))$$

the smallest projective linear subspace of $\mathbb{P}(V)$ containing U.

Theorem 6.2.4 (Dimemsion Theorem) Let $E, F \subset \mathbb{P}^n$ be projective linear subspaces. Then

$$\dim(E \cap F) = \dim(E) + \dim(F) - \dim\langle E, F \rangle$$

Proof Let ■

Theorem 6.2.5 Any two distinct lines L_1 and L_2 in $\mathbb{P}^2$ intersect in a point.

Proof Note that L_1 and L_2 are projections of two planes A_1, A_2 through the origin. Since A_1, A_2 have dimension 2 and are distinct. Thus their intersection must have dimension 1 and their span has dimension 3. But $\langle L_1, L_2\rangle = \mathbb{P}(\text{span}(A_1, A_2)) = \mathbb{P}(\mathbb{R}^3)$ and thus $\dim(\langle L_1, L_2\rangle) = 2$. By the above theorem, we must have $\dim(L_1 \cap L_2) = 0$ and thus it is exactly a point. ■

6.3 Projective Transformations

Definition 6.3.1 (Projective Transformations) A projective transformation of $\mathbb{P}^n$ is a map $T|_A : \mathbb{P}^n \to \mathbb{P}^n$ defined by

$$T([x]) = [Ax]$$

where $A \in \text{GL}(n+1, \mathbb{R})$.

Definition 6.3.2 (Projective General Linear Group) The projective general linear group of a vector space V over k with dimension n is the group of all invertible linear transformations unique up to scalar multiplication. That is, we say that $T_1 \sim T_2$ if $T_1 = \lambda T_2$ for some $\lambda \in k$ and $\lambda \neq 0$. It is denoted as $\text{PGL}(n+1, k)$.

Definition 6.3.3 (Projective Frame of Reference) A projective frame of reference for $\mathbb{P}^n$ is an ordered set of $n+2$ points, $P_0, \dots, P_{n+1} \in \mathbb{P}^n$ such that any $n+1$ points are linearly independent. The standard frame of reference is just $[e_1], \dots, [e_{n+1}], [e_1 + \cdots + e_{n+1}]$.

Proposition 6.3.4 There is a bijection between projective transformations of $\mathbb{P}^n$ and projective frames of references of $\mathbb{P}^n$ as follows:

$$\phi(T) = \{T([e_1]), \dots, T([e_{n+1}]), T([e_1 + \cdots + e_{n+1}])\}$$

where T is the projective transformation.

This means that specifying any $n+2$ points in $\mathbb{P}^n$ such that they form a projective frame of reference induces a unique projective transformation.

6.4 Perspectivities

Definition 6.4.1 (Perspectivities) Let Π_1, Π_2 be two hyperplanes in $\mathbb{P}^n$. A perspectivity $f : \Pi_1 \to \Pi_2$ from a point $O \in \mathbb{P}^n$ but not in Π_1 or Π_2 is a map given by

$$f(P) = \Pi_2 \cap \langle O, P\rangle$$

Lemma 6.4.2 Perspectivities are well defined.

Proof We want to show that $f(P)$ is a point. We appeal to the dimension theorem. We have that

$$\dim(\Pi_2 \cap \langle O, P\rangle) = \dim(O) + \dim(\Pi_2) - \dim(\langle \Pi_2 \langle O, P\rangle\rangle)$$

Since O is necessarily not in Π_1, we have $\langle O, P\rangle$ is a projective line. Moreover, since O is not in Π_2, we must have that the span of the hyperplane Π_2 and a line $\langle O, P\rangle$ must be the projective space itself. Thus we have that

$$\dim(\Pi_2 \cap \langle O, P\rangle) = 0$$

This means that $f(P)$ must be a point in projective space. ■

Definition 6.4.3 (Cross Ratio) Let P, Q, R, S be distinct and ordered points on a projective line in $\mathbb{P}^n$. Define the cross ratio between the four projective points to be the ratio

$$(P, Q; R, S) = \left(\frac{p-r}{p-s}\right)\left(\frac{q-s}{q-r}\right)$$

where the ratio between vectors lying on the same projective line is defined to be the λ in $\frac{\lambda v}{v}$.

Proposition 6.4.4 Projective linear maps perserve perspectivity.

Proof By linearity, we must have

$$\frac{T(\lambda v)}{T(v)} = \frac{\lambda v}{v}$$

■

Lemma 6.4.5 Perspectivities are projective linear maps.

6.5 Important Theorems

Definition 6.5.1 (Triangles in Projective Space) Let $P, Q, R \in \mathbb{P}^n$ be distinct. The triangle $\triangle PQR$ in $\mathbb{P}^n$ is defined to be the three points and the three sides spanned by three pair of points.

Definition 6.5.2 (In Perspective from a Point) Two triangles $\triangle PQR$ and $\triangle P'Q'R'$ in $\mathbb{P}^n$ are said to be in persepective from a point O if the lines passing through PP', QQ' and RR' intersect at a O.

Definition 6.5.3 (In Perspective from a Line) Two triangles $\triangle PQR$ and $\triangle P'Q'R'$ in $\mathbb{P}^n$ are said to be in persepective from a line L if the points $PQ \cap P'Q'$, $QR \cap Q'R'$ and $PR \cap P'R'$ are collinear.

Note that this definition makes sense since any two lines in projective space must meet at one point.

Theorem 6.5.4 (Desargues' Theorem) If $\triangle PQR$ and $\triangle P'Q'R'$ are two distinct triangles in $\mathbb{P}^n$ which are in perspective from a point, then they are also in perspective from a line.

Theorem 6.5.5 (Pappus' Theorem) Let L and L' be distinct projective lines in $\mathbb{P}^2$. Let P, Q, R in L but not in L' and P', Q', R' in L' but not in L all be distinct points. Then the intersection points $PQ' \cap P'Q$, $PR' \cap P'R$ and $QR' \cap Q'R$ are collinear.

6.6 Axiomatic Projective Geometry

Just as Euclidean geometry has 5 axioms to follow, we can also establish axioms for projective geometry.

Definition 6.6.1 (Axiomatic Projective Geometry) An axiomatic projective plane (P, L, I) consists of a set P called the set of points, a set L, the set of lines and a relation $I \subseteq P \times L$, the incidence relation. For $l_1, l_2 \in L$, define

$$l_1 \cap l_2 = \{p \in P | p \in l_1, p \in l_2\}$$

These sets must satisfy the following four axioms:

1. Every line contains at least three distinct points:

$$\forall l \in L \exists \text{ distinct } x, y, z \in P \text{ such that } (x, l) \in I, (y, l) \in I, (z, l) \in I$$

2. Every point is contained in at least three distinct lines:

$$\forall x \in P \exists \text{ distinct } l, m, n \in L \text{ such that } (x, l) \in I, (x, m) \in I, (x, n) \in I$$

3. Any two points span a unique line:

$$\forall x \neq y \in P \exists ! l \in L \text{ such that } (x,l) \in I, (y,l) \in I$$

4. Any two distinct lines intesect at a unique point:

$$\forall l \neq m \in I \exists ! x \in P \text{ such that } (x,l) \in I, (x,m) \in I$$